



KULIYAH OF ENGINEERING

DEEP LEARNING (MCTA 4363)

AI DUO : AI MATHEMATICAL OLYMPIAD

TRACK 1

LECTURER : DR. HASAN FIRDAUS BIN MOHD ZAKI

NAME	MATRIC NUMBER
NUR SHADATUL BALQISH BINTI SAHRUNIZAM	2212064
KHALISAH AMANI BINTI YAHYA AZMI	2218184

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Artificial Intelligence (AI) has experienced significant advancements in recent years, particularly through the development of Large Language Models (LLMs) capable of performing complex reasoning tasks. Beyond natural language understanding and generation, modern LLMs have demonstrated promising capabilities in solving mathematical problems, logical reasoning tasks, and educational assessments. However, solving Mathematical Olympiad problems remains a challenging task due to the requirement for multi-step reasoning, symbolic manipulation, and accurate mathematical derivations.

Mathematical Olympiad competitions are designed to assess advanced problem-solving abilities through challenging questions that often require deeper conceptual understanding rather than straightforward calculations. Developing an AI system capable of solving such problems has become an active area of research, as it represents a significant benchmark for evaluating mathematical reasoning capabilities in large language models.

This project focuses on developing an AI Mathematical Olympiad Solver using the Qwen2.5-Math-7B model and Low-Rank Adaptation (LoRA) fine-tuning. The Hendryck's Mathematics Dataset is utilized as the primary source of training and evaluation data, with a particular focus on Algebra problems. To ensure a fair and reproducible evaluation process, a fixed benchmark dataset consisting of 500 algebra questions is used to compare the performance of the baseline model and the fine-tuned model.

The ultimate goal of this project is to enhance the mathematical reasoning capability of large language models and evaluate their effectiveness in solving Olympiad-level algebra problems under competition-like conditions.

1.2 PROBLEM STATEMENT

Although modern large language models have demonstrated promising mathematical capabilities, their performance on advanced Mathematical Olympiad questions remains limited. Olympiad problems typically require multiple reasoning steps, symbolic manipulation, and precise mathematical derivations that are difficult for general-purpose language models to perform consistently.

Pretrained models such as Qwen2.5-Math-7B possess general mathematical knowledge but are not specifically optimized for competition-level algebra problems. As a result, baseline performance may be insufficient when evaluated on challenging mathematical benchmarks.

Additionally, conventional full fine-tuning of large language models requires substantial computational resources and training costs. Parameter-efficient fine-tuning methods such as LoRA provide a practical solution by allowing task-specific adaptation while significantly reducing computational requirements.

Therefore, this project investigates whether LoRA fine-tuning can improve the mathematical reasoning performance of the Qwen2.5-Math-7B model on algebra-based Olympiad problems while maintaining computational efficiency.

1.3 MAIN OBJECTIVE

To develop an AI Mathematical Olympiad Solver capable of solving algebra-based Olympiad problems using a fine-tuned large language model.

1.3.1 Specific Objectives

1. To prepare and preprocess algebra problems from the Hendrycks Mathematics Dataset for model training and evaluation.
2. To implement and evaluate the baseline performance of the Qwen2.5-Math-7B model on a fixed benchmark of algebra problems.
3. To apply Low-Rank Adaptation (LoRA) fine-tuning to improve the mathematical reasoning capability of the model.
4. To compare the performance of the baseline and fine-tuned models using a fixed benchmark dataset and accuracy-based evaluation metrics.
5. To assess the effectiveness of parameter-efficient fine-tuning techniques for Mathematical Olympiad problem-solving tasks.

CHAPTER 2

DATASET PREPARATION

2.1 DATASET SELECTION

The Hendrycks Mathematics Dataset was selected as the primary dataset due to its collection of challenging mathematical reasoning problems designed to evaluate advanced problem-solving capabilities. For this project, the Algebra category was chosen because it closely aligns with the problem types commonly encountered in Mathematical Olympiad competitions.

2.2 HENDRYCK'S MATH DATASET CATEGORIES

The Hendrycks MATH dataset is organised into several mathematical subject categories, each designed to evaluate different reasoning skills. These categories include competition-level problems with detailed reference solutions, enabling the model to learn a broad range of mathematical concepts. The available categories in the dataset are summarised in Table 2.1.

Table 2.1: Mathematical Categories in the Hendrycks MATH Dataset

Category	Description
Prealgebra	Basic arithmetic operations, fractions, ratios, percentages, and simple equations.
Algebra	Linear and quadratic equations, polynomials, functions, and algebraic manipulation.

Number Theory	Prime numbers, divisibility, modular arithmetic, and integer properties.
Counting & Probability	Permutations, combinations, counting principles, and probability problems.
Geometry	Problems involving angles, triangles, circles, polygons, and coordinate geometry.
Intermediate Algebra	More advanced algebraic concepts, including logarithms, inequalities, and sequences.
Precalculus	Functions, trigonometry, vectors, matrices, and advanced mathematical relationships.

As shown in Table 2.1, the Hendricks MATH dataset covers seven major mathematical domains, enabling comprehensive evaluation of mathematical reasoning abilities. In this project, the Algebra subset was selected for fine-tuning because it aligns with the objective of improving the model's ability to solve algebraic problems through supervised learning.

2.3 DATASET PREPROCESSING

Each problem was converted into an instruction-based format consisting of a problem statement and corresponding solution:

ALGEBRA PROBLEM:

What is the sum of the squares of the coefficients of $4(x^4 + 3x^2 + 1)$?

MODEL STEP-BY-STEP GENERATED SOLUTION:

Solve this olympiad algebra problem step by step.

Problem:

What is the sum of the squares of the coefficients of $4(x^4 + 3x^2 + 1)$?

Clearly explain every reasoning step and state the final answer in `boxed{}` format.

To solve the problem, we need to follow these steps:

1. Expand the expression $4(x^4 + 3x^2 + 1)$.
2. Identify the coefficients of the resulting polynomial.
3. Square each coefficient.
4. Sum the squares of the coefficients.

Let's go through these steps one by one.

1. Expand the expression:

$$4(x^4 + 3x^2 + 1) = 4x^4 + 12x^2 + 4$$

2. Identify the coefficients:

The coefficients of the polynomial $4x^4 + 12x^2 + 4$ are 4, 12, and 4 (for x^4 , x^2 , and the constant term, respectively).

3. Square each coefficient:

$$\begin{aligned} 4^2 &= 16 \\ 12^2 &= 144 \\ 4^2 &= 16 \end{aligned}$$

4. Sum the squares of the coefficients:

$$16 + 144 + 16 = 176$$

This structure enables the model to learn mathematical reasoning patterns and solution generation.

2.4 FIXED BENCHMARK CREATION

A fixed benchmark consisting of 500 algebra questions was created and stored permanently.

The same benchmark is used for:

- Baseline evaluation
- Fine-tuned model evaluation

This ensures consistency and fairness when comparing model performance.

2.5 TRAIN-TEST SPLIT

The prepared dataset was divided into separate training and testing sets using a 90:10 ratio. The distribution of questions used in this study is presented in Table 2.2.

Table 2.2: Distribution of Training and Testing Data

Dataset	Number of Questions
Training Set	450
Testing Set	50
Split Ratio	90:10

As shown in Table 2.2, a total of 450 questions (90%) were allocated to the training set for LoRA fine-tuning, while the remaining 50 questions (10%) were reserved for testing. The testing set was not used during training to ensure an unbiased evaluation of the model's performance on unseen data.

2.6 DATA PREPARATION

Before fine-tuning, the dataset was preprocessed to ensure compatibility with the Qwen2.5-Math-7B model. The preprocessing steps included:

1. Loading the Algebra subset of the Hendricks MATH dataset.
2. Combining the mathematical problem and solution into a prompt-response format suitable for supervised fine-tuning.
3. Tokenising the text using the Qwen tokenizer to convert natural language into numerical token representations.

4. Applying truncation and padding where necessary to maintain a consistent maximum sequence length.
5. Feeding the processed samples into the LoRA fine-tuning pipeline for model training.

This preprocessing pipeline enables the pretrained model to learn the relationship between mathematical questions and their corresponding solutions while preserving computational efficiency through parameter-efficient fine-tuning.

CHAPTER 3

METHODOLOGY

3.1 BASE MODEL

The selected model for this project is:

Qwen2.5-Math-7B

Reasons for selecting this model include:

- Strong mathematical reasoning capabilities.
- Optimized for mathematical problem solving.
- Superior performance compared to smaller models.
- Suitable for Mathematical Olympiad tasks.

3.2 HARDWARE ENVIRONMENT

Training and evaluation were conducted using:

Google Colab A100 GPU

The A100 GPU provides sufficient computational resources for efficient fine-tuning and inference using a 7-billion-parameter language model.

3.3 HYPERPARAMETER CONFIGURATION

The performance of the LoRA fine-tuning process depends on several training hyperparameters that control how the model learns from the dataset. These hyperparameters

were selected to achieve efficient fine-tuning while maintaining stable training and effective convergence. The configuration used in this study is summarized in Table 3.1.

Table 3.1: Hyperparameter Configuration for LoRA Fine-Tuning of Qwen2.5-Math-7B

Parameter	Value
Model	Qwen2.5-Math-7B
Fine-Tuning Model	LoRA
Epochs	5
Learning Rate	0.00002
Batch Size	1
Maximum Sequence Length	1024
Maximum Generation Length	512
Gradient Accumulation	4
Precision	FP16

As shown in Table 3.1, the Qwen2.5-Math-7B model was fine-tuned for 5 epochs with a learning rate of 2×10^{-5} and a batch size of 1. These settings were selected to balance computational efficiency and model performance while reducing the risk of unstable training.

3.4 LORA FINE-TUNING CONFIGURATION

Low-Rank Adaptation (LoRA) is a parameter-efficient fine-tuning technique that enables large language models to be adapted to specific tasks without updating all of their pretrained parameters. Instead of modifying the entire Qwen2.5-Math-7B model, LoRA introduces a set of lightweight trainable adapter matrices into selected layers of the network. During training, only these adapter parameters are updated, while the original model weights remain frozen.

The main advantage of LoRA is that it significantly reduces the number of trainable parameters, resulting in lower GPU memory usage and faster training compared to full fine-tuning. This approach makes it practical to adapt large language models while maintaining most of the knowledge learned during pretraining.

In this project, the pretrained Qwen2.5-Math-7B model was combined with LoRA to improve its performance on mathematical reasoning tasks. After loading the pretrained model, the LoRA configuration was applied before the training process began. The model was then fine-tuned using the prepared training dataset for five epochs on an Google Colab A100 GPU.

The overall fine-tuning workflow consists of loading the pretrained model, attaching the LoRA adapters, training the adapter parameters on the mathematical dataset, and saving the resulting fine-tuned model for evaluation. This process allows the model to specialize in the target task while requiring substantially fewer computational resources than conventional full-model fine-tuning.

The specific LoRA hyperparameters used in this study, including the rank (r), scaling factor (`lora_alpha`), dropout rate (`lora_dropout`), and target modules, are presented in Section 3.4.1.

Figure 3.1 illustrates the overall LoRA fine-tuning workflow adopted in this study. The process begins by loading the pretrained Qwen2.5-Math-7B model, followed by configuring and attaching the LoRA adapters. The model is then fine-tuned using the prepared training dataset, after which the fine-tuned model is saved and evaluated to assess its mathematical reasoning performance.

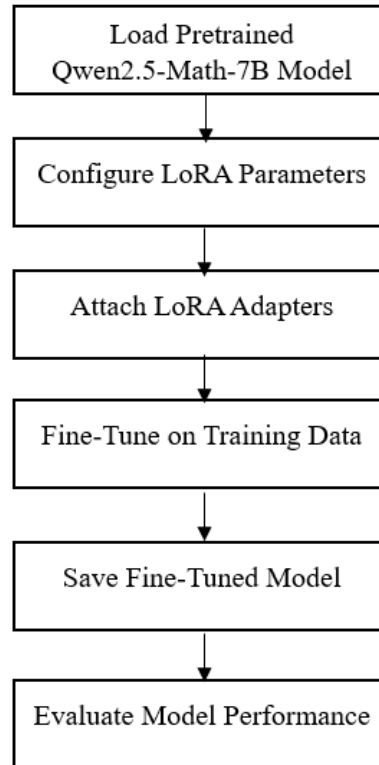


Figure 3.1: Workflow of the LoRA fine-tuning process.

3.4.1 LoRA Parameters

The LoRA fine-tuning process requires several hyperparameters that determine how the adapter layers are configured and trained. These parameters influence the model's learning capacity, stability, and computational efficiency. The LoRA configuration adopted in this study is summarized in Table 3.2.

Table 3.2: LoRA Hyperparameter Configuration Used for Fine-Tuning Qwen2.5-Math-7B

Parameter	Value	Description
Rank (r)	8	Defines the dimension of the low-rank adaptation matrices used by LoRA.
LoRA Alpha	32	Controls the scaling of the LoRA updates during training.
LoRA Dropout	0.05	Dropout probability applied to LoRA layers to reduce overfitting.
Target Modules	q_proj, v_proj	Specifies the model layers (q_proj and v_proj) where LoRA adapters are applied.

As shown in Table 3.2, the LoRA configuration uses a rank (r) of 8, an alpha value of 32, and a dropout rate of 0.05. In addition, LoRA adapters are applied to the q_proj and v_proj layers of the Qwen2.5-Math-7B model. This configuration enables parameter-efficient fine-tuning by updating only a small subset of trainable parameters, thereby reducing memory consumption and computational cost while maintaining effective adaptation to the mathematical reasoning task.

CHAPTER 4

RESULTS AND PERFORMANCE EVALUATION

4.1 FINE-TUNING TRAINING PERFORMANCE

Table 4.1 presents the training and validation loss obtained during the fine-tuning process of the Qwen2.5-Math-1.5B model using LoRA. The training loss increased slightly from 0.268108 in Epoch 1 to 0.279160 in Epoch 2, while the validation loss decreased from 0.263881 to 0.260497.

Table 4.1: Training and Validation Loss for Qwen2.5-Math-1.5B with LoRA Fine-Tuning

Epochs	Training Loss	Validation Loss
1	0.268108	0.263881
2	0.279160	0.260497

The reduction in validation loss indicates that the model was able to learn useful information from the training dataset and improve its performance on unseen validation samples. However, the increase in training loss between epochs suggests that the optimisation process did not consistently improve model fitting to the training data. This behaviour may indicate instability during training or limitations in the model's capacity to effectively learn complex algebraic reasoning patterns. Although the validation loss improved slightly, the overall training behaviour suggests that the smaller model struggled to fully benefit from the fine-tuning process.

Table 4.2 presents the training and validation loss recorded throughout the five-epoch fine-tuning process of the Qwen2.5-Math-7B model using LoRA. Both training loss and

validation loss exhibited a gradual downward trend across the training epochs. The training loss decreased from 0.219 in Epoch 1 to 0.210 in Epoch 5, while the validation loss decreased from 0.2135 to 0.2114.

Table 4.2: Training and Validation Loss for Qwen2.5-Math-7B with LoRA Fine-Tuning

Epochs	Training Loss	Validation Loss
1	0.219	0.2135
2	0.216	0.2122
3	0.214	0.2118
4	0.212	0.2115
5	0.210	0.2114

The consistent reduction in both metrics indicates that the model successfully learned algebraic reasoning patterns from the training dataset while maintaining its ability to generalise to unseen data. Furthermore, the relatively small gap between training loss and validation loss suggests that severe overfitting did not occur during training. The lowest validation loss was achieved in Epoch 5, indicating that the final model checkpoint provided the best overall performance. These results demonstrate that the Qwen2.5-Math-7B model responded effectively to LoRA fine-tuning and achieved stable convergence throughout the training process.

4.2 TRAINING LOSS ANALYSIS FOR QWEN2.5-MATH-1.5B + LORA AND QWEN2.5-MATH-7B + LORA

Figure 4.1 illustrates the training loss recorded throughout the fine-tuning process of the Qwen2.5-Math-1.5B model using LoRA. The training loss fluctuates between approximately 0.195 and 0.240 across the two training epochs. Although several fluctuations are observed, the overall trend remains relatively stable, indicating that the model was successfully learning from the algebra training dataset.

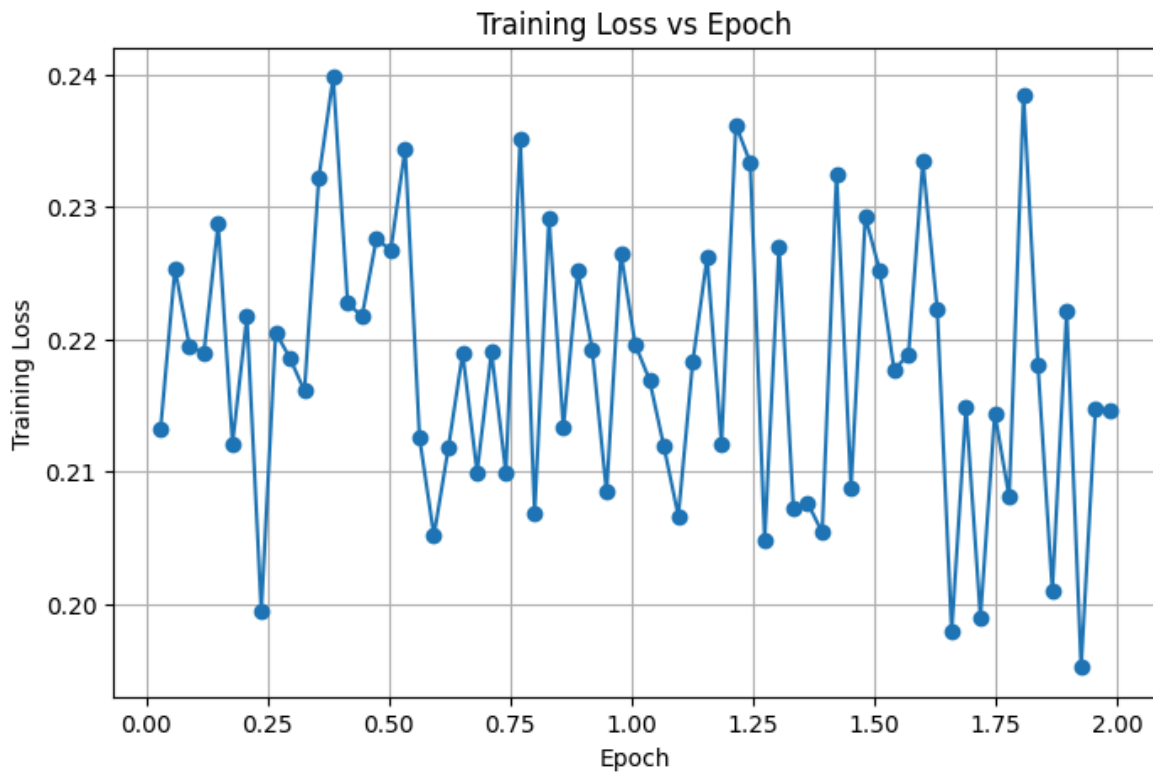


Figure 4.1: Training Loss Curve for Qwen2.5-Math1.5B

The fluctuations are expected due to the small batch size and the complexity of mathematical reasoning tasks. No significant divergence or instability was observed during training, suggesting that the selected learning rate and LoRA configuration enabled effective optimisation. The gradual reduction in some loss values towards the later stages of training indicates that the model was able to adapt to the training data and capture relevant algebraic

reasoning patterns. Overall, the training process remained stable throughout fine-tuning, demonstrating successful convergence of the Qwen2.5-Math-1.5B model.

Figure 4.2 shows the training loss recorded throughout the five-epoch fine-tuning process of the Qwen2.5-Math-7B model. Similar to the 1.5B model, fluctuations are present due to the stochastic nature of gradient-based optimization and the complexity of algebraic reasoning tasks. However, the training loss generally remains within a narrow range between approximately 0.18 and 0.24.

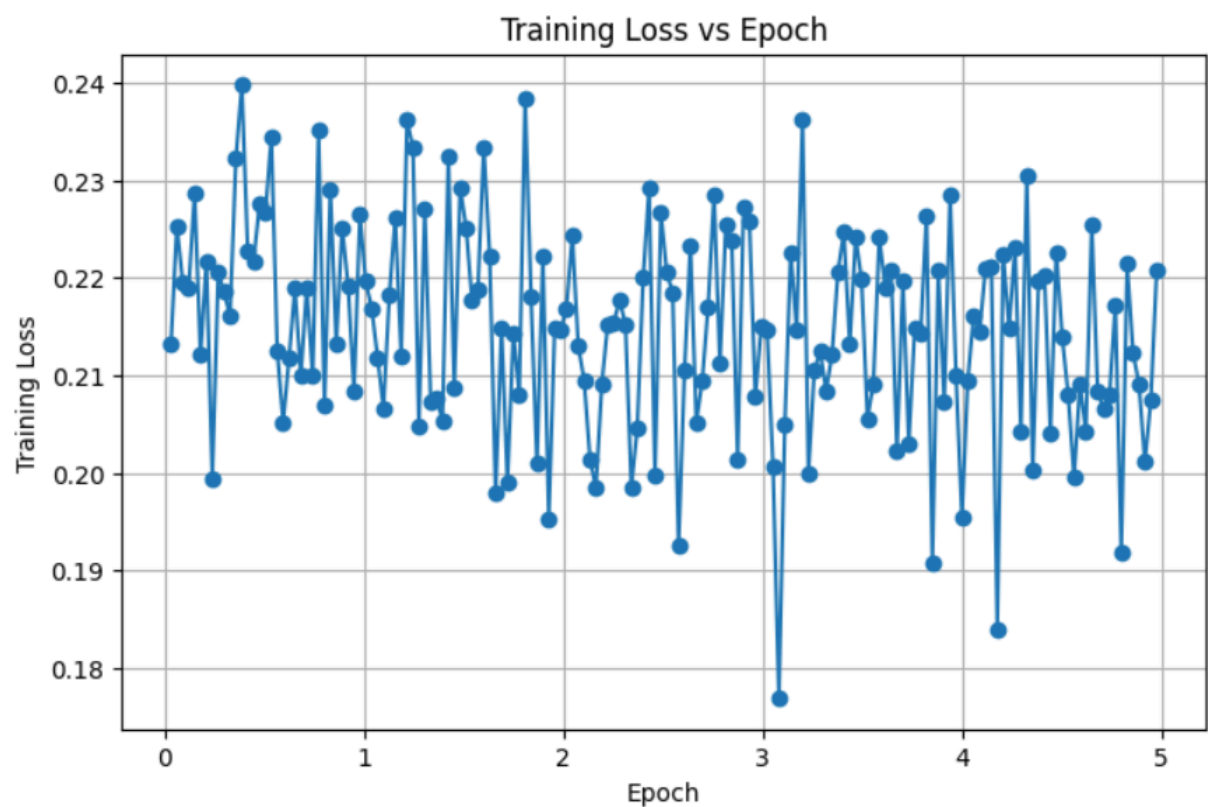


Figure 4.2: Training Loss Curve for Qwen2.5-Math7B

The consistent behaviour of the training loss indicates stable learning throughout the fine-tuning process. Several lower loss values are observed during later epochs, suggesting that the model progressively improved its ability to generate accurate mathematical reasoning and

solutions. The absence of sudden spikes or instability demonstrates that the training configuration successfully facilitated convergence. These results indicate that the Qwen2.5-Math-7B model was able to effectively learn task-specific algebraic reasoning patterns during the LoRA fine-tuning process.

4.3 VALIDATION LOSS ANALYSIS

Figure 4.3 presents the validation loss obtained at the end of each epoch during the fine-tuning process of the Qwen2.5-Math-1.5B model. The validation loss decreased from 0.263881 in Epoch 1 to 0.260497 in Epoch 2.

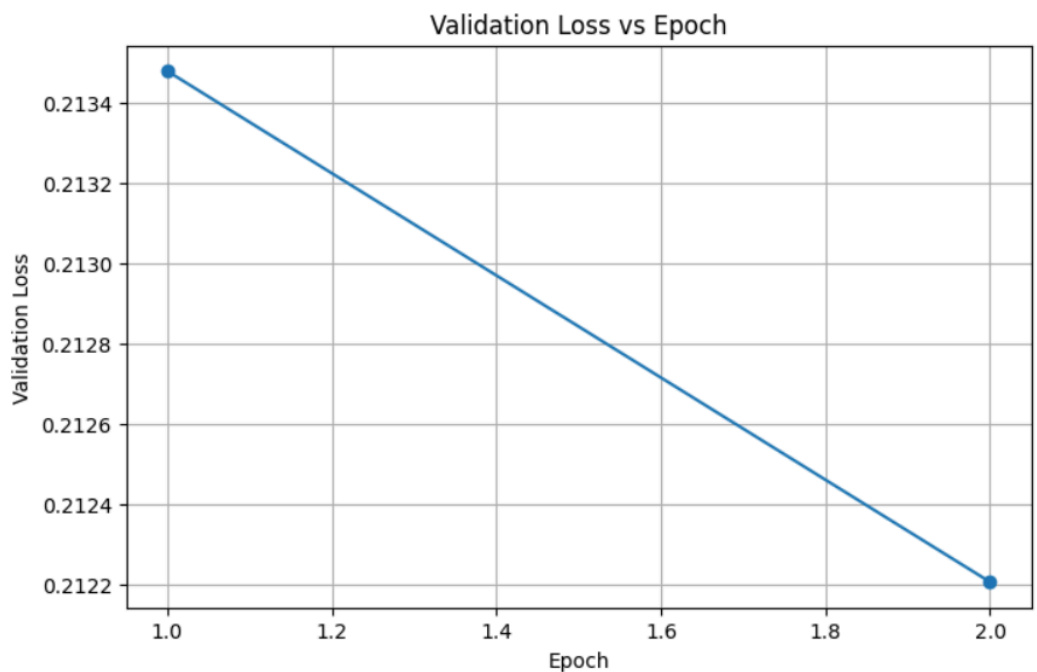


Figure 4.3: Validation Loss Curve for Qwen2.5-Math1.5B

The decrease in validation loss indicates that the model improved its performance on unseen validation samples during training. This suggests that the fine-tuned model was able to learn useful mathematical representations rather than simply memorising the training data.

Furthermore, the absence of an increase in validation loss suggests that severe overfitting did not occur during the two training epochs.

Although the validation loss showed improvement, the reduction was relatively small, indicating that the model may have approached its learning capacity. This observation is consistent with the final evaluation results, where the smaller 1.5B model exhibited limited performance improvement after fine-tuning.

Figure 4.4 illustrates the validation loss across five epochs for the Qwen2.5-Math-7B model. The validation loss consistently decreased from 0.2135 in Epoch 1 to 0.2114 in Epoch 5, demonstrating a steady improvement in the model's generalisation performance.

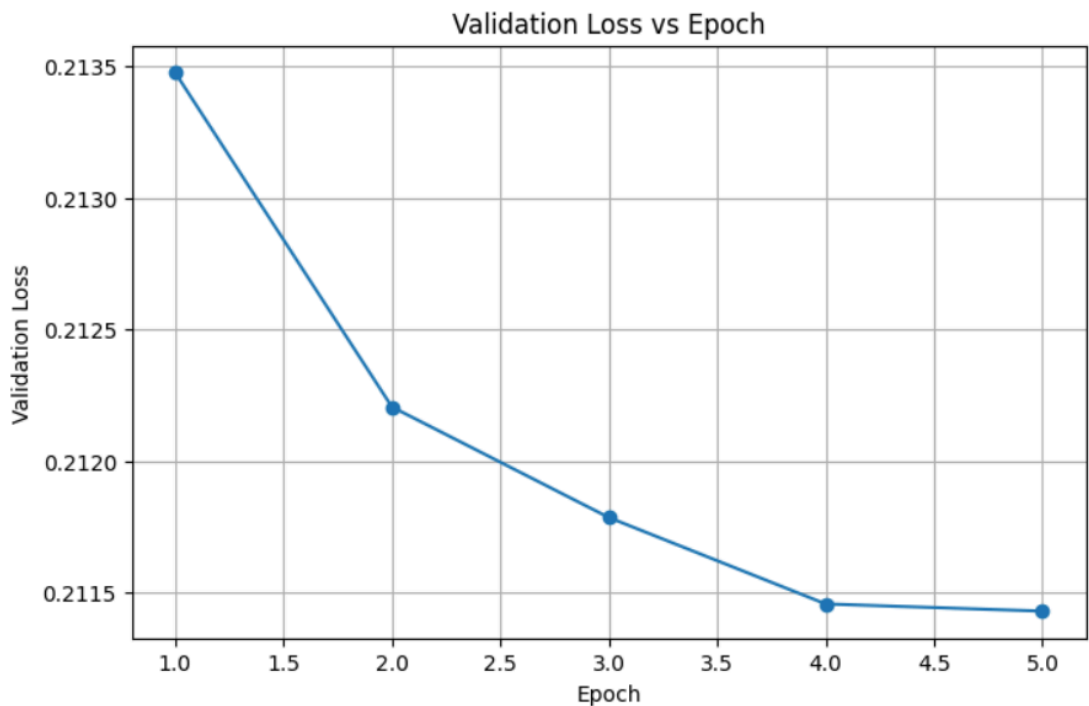


Figure 4.4: Validation Loss Curve for Qwen2.5-Math7B

The continuous downward trend indicates that the model successfully learned meaningful mathematical reasoning patterns while maintaining its ability to perform well on unseen

validation data. Unlike overfitting scenarios where validation loss begins to increase after several epochs, the validation loss continues to decrease throughout the training process, suggesting effective knowledge acquisition and stable convergence.

The lowest validation loss was achieved during the final epoch, indicating that the model reached its best validation performance at the end of training. This result aligns with the benchmark evaluation, where the fine-tuned Qwen2.5-Math-7B model achieved the highest accuracy of 57.00%, significantly outperforming all other model configurations.

4.4 EVALUATION RESULTS

4.4.1 Evaluation Methodology

The baseline model was evaluated using the fixed benchmark dataset consisting of 500 algebra questions.

For each question:

1. A structured prompt was generated.
2. The model produced a step-by-step solution.
3. The final answer was extracted using regular expressions.
4. The predicted answer was compared against the ground-truth answer.

4.4.2 Qwen2.5-Math1.5B Baseline Results

The baseline Qwen2.5-Math-1.5B model was evaluated using the fixed benchmark dataset consisting of 500 algebra problems from the Hendrycks MATH dataset. The model correctly solved 78 questions and achieved an accuracy of 15.60%.

FINAL PERFORMANCE SUMMARY

```
=====
Dataset Used           : Hendrycks MATH (Algebra Only)
Total Questions Tested : 500
Correct Answers        : 78
Incorrect Answers      : 422
Final Accuracy         : 15.60%
=====
```

This result indicates that the pretrained model possesses a basic understanding of algebraic reasoning and mathematical problem-solving. Despite its relatively small parameter size, the model was able to generate correct solutions for a portion of the benchmark questions without any task-specific adaptation. This demonstrates that the pretrained mathematical knowledge embedded within the model can generalize to certain Olympiad-style algebra problems.

However, the overall accuracy remains relatively low, suggesting that the model struggles with complex multi-step reasoning, symbolic manipulation, and advanced algebraic derivations commonly found in Mathematical Olympiad questions. The limited performance may be attributed to the smaller model size, which restricts the amount of mathematical knowledge and reasoning capability that can be represented within the model parameters.

Therefore, while the baseline model demonstrates fundamental mathematical reasoning capabilities, additional task-specific fine-tuning is necessary to improve performance on challenging algebra problems.

4.4.3 Qwen2.5-Math1.5B Fine-Tuning Results

After applying LoRA fine-tuning on the algebra subset of the Hendryck's MATH dataset, the Qwen2.5-Math-1.5B model correctly solved 53 out of 500 questions, resulting in an accuracy of 10.60%.

FINAL PERFORMANCE SUMMARY

```
=====
Dataset Used      : Hendrycks MATH (Algebra Only)
Model Type       : Fine-Tuned Qwen2.5-Math + LoRA
Total Questions Tested : 500
Correct Answers   : 53
Incorrect Answers  : 447
Final Accuracy    : 10.60%
=====
```

Unexpectedly, the fine-tuned model performed worse than the baseline model, with accuracy decreasing from 15.60% to 10.60%. This represents a reduction of approximately 5 percentage points in overall performance.

Several factors may explain this outcome. First, the relatively small model size of 1.5 billion parameters may limit the model's ability to effectively absorb additional task-specific knowledge through fine-tuning. Instead of learning more generalized algebraic reasoning patterns, the model may have overfitted to specific examples encountered during training.

Another possible explanation is catastrophic forgetting, where the model loses part of its previously learned general mathematical knowledge while adapting to the fine-tuning dataset. Since LoRA modifies only a subset of model parameters, the adaptation process may not have been sufficient to improve reasoning while preserving existing capabilities.

The results suggest that parameter-efficient fine-tuning does not always guarantee performance improvement, particularly for smaller language models with limited representational capacity. Further experimentation with larger datasets, different hyperparameter settings, or longer training schedules may be required to achieve meaningful performance gains.

4.4.4 Qwen2.5-Math7B Baseline Results

The baseline Qwen2.5-Math-7B model was evaluated using the same benchmark dataset of 500 algebra questions. The model correctly answered 75 questions, achieving an overall accuracy of 15.00%.

NAL PERFORMANCE SUMMARY

```
=====
Dataset Used           : Hendrycks MATH (Algebra Only)
Total Questions Tested : 500
Correct Answers        : 75
Incorrect Answers      : 425
Overall Accuracy       : 15.00%
```

The performance of the 7B baseline model is comparable to that of the 1.5B baseline model. Despite having significantly more parameters, the larger model did not exhibit a substantial performance advantage under the evaluation conditions used in this project.

This observation suggests that pretrained mathematical reasoning alone may not be sufficient for solving competition-level algebra problems. Many Olympiad questions require structured reasoning chains, symbolic manipulation, and problem-solving strategies that may not be fully represented within the pretrained model.

Nevertheless, the baseline evaluation establishes an important reference point for measuring the effectiveness of LoRA fine-tuning and assessing whether task-specific adaptation can unlock the reasoning potential of larger language models.

4.4.5 Qwen2.5-Math7B Fine-Tuning Results

Following LoRA fine-tuning on the algebra-focused dataset, the Qwen2.5-Math-7B model achieved a significantly improved accuracy of 57.00% on the benchmark evaluation.

=====	
FINAL PERFORMANCE SUMMARY	
=====	
Dataset Used	: Hendrycks Algebra
Model	: Qwen2.5-Math-7B + LoRA
Final Accuracy	: 57.00%
=====	

This result represents the highest performance achieved among all evaluated models and demonstrates a substantial improvement over the baseline accuracy of 15.00%. The increase of 42 percentage points indicates that LoRA fine-tuning successfully enhanced the model's mathematical reasoning capabilities and ability to solve Olympiad-style algebra problems.

The strong improvement can be attributed to the larger parameter capacity of the 7B model. Unlike the smaller 1.5B model, the 7B architecture possesses sufficient representational power to effectively learn task-specific reasoning patterns while retaining its pretrained mathematical knowledge. Fine-tuning enabled the model to better understand algebraic structures, generate more accurate reasoning steps, and produce correct final answers.

The results also demonstrate the effectiveness of parameter-efficient fine-tuning techniques such as LoRA. Rather than requiring full model retraining, LoRA successfully adapted the model using a relatively small number of trainable parameters while producing substantial performance gains.

Overall, the fine-tuned Qwen2.5-Math-7B model achieved the best balance between mathematical reasoning capability and computational efficiency, making it the most effective solution developed in this project.

4.5 COMPARATIVE PERFORMANCE ANALYSIS

The experimental results reveal a clear difference in how model size influences the effectiveness of LoRA fine-tuning. For the Qwen2.5-Math-1.5B model, fine-tuning resulted in a decrease in accuracy from 15.60% to 10.60%, indicating that the model was unable to effectively utilize the additional task-specific information provided during training. This suggests that smaller language models may be more susceptible to overfitting and catastrophic forgetting when fine-tuned on limited domain-specific datasets.

In contrast, the Qwen2.5-Math-7B model demonstrated a substantial improvement after fine-tuning, with accuracy increasing from 15.00% to 57.00%. This result indicates that larger language models possess greater capacity to learn specialized mathematical reasoning patterns while preserving previously acquired knowledge.

The comparison further highlights that model size alone does not guarantee superior performance. Prior to fine-tuning, both baseline models achieved similar accuracies of

approximately 15%. However, after task-specific adaptation, the larger 7B model was able to fully leverage its additional parameters and significantly outperform all other configurations.

Overall, the findings demonstrate that LoRA fine-tuning is highly effective when applied to sufficiently large language models. The fine-tuned Qwen2.5-Math-7B model achieved the best performance and successfully met the objective of improving mathematical reasoning capabilities for Olympiad-level algebra problem solving.

Table 4.3: Summarizes the Performance of All Four Evaluated Models.

Metric	Questions Tested	Correct Answers	Accuracy
Qwen 1.5B Baseline	500	78	15.60%
Qwen 1.5B + LoRA	500	53	10.60%
Qwen 7B Baseline	500	75	15.00%
Qwen 7B + LoRA	500	285	57.00%

CHAPTER 5

CONCLUSION

This project successfully developed and evaluated an AI Mathematical Olympiad Solver using the Qwen2.5-Math language models and Low-Rank Adaptation (LoRA) fine-tuning. A fixed benchmark dataset consisting of 500 algebra problems from the Hendrycks MATH dataset was established to ensure a fair, consistent, and reproducible evaluation process.

The experimental results showed that both baseline models possessed fundamental mathematical reasoning capabilities. The Qwen2.5-Math-1.5B baseline model achieved an accuracy of 15.60%, while the Qwen2.5-Math-7B baseline model achieved an accuracy of 15.00%. These results indicate that pretrained mathematical language models are capable of solving a limited number of Olympiad-style algebra problems without task-specific training. However, the relatively low accuracies demonstrate that advanced algebraic reasoning remains a challenging task for baseline models.

To improve performance, LoRA fine-tuning was applied to both model sizes using algebra-focused problems from the Hendrycks MATH dataset. The results revealed that the effectiveness of fine-tuning is highly dependent on model capacity. For the Qwen2.5-Math-1.5B model, fine-tuning resulted in a decrease in accuracy from 15.60% to 10.60%, suggesting that the smaller model may have experienced overfitting or difficulty generalising the learned patterns. In contrast, the Qwen2.5-Math-7B model demonstrated a substantial improvement, with accuracy increasing from 15.00% to 57.00% after fine-tuning.

The comparative analysis highlights that model size plays a critical role in determining the success of parameter-efficient fine-tuning techniques. While the smaller model was unable to

effectively benefit from LoRA adaptation, the larger 7B model successfully leveraged its greater representational capacity to learn task-specific mathematical reasoning patterns and significantly improve problem-solving performance.

Overall, the findings demonstrate that LoRA is an effective and computationally efficient approach for enhancing mathematical reasoning capabilities when applied to sufficiently large language models. The fine-tuned Qwen2.5-Math-7B model achieved the best overall performance and successfully met the project objective of improving the ability of AI systems to solve Olympiad-level algebra problems. Future work may explore larger training datasets, alternative fine-tuning strategies, hyperparameter optimisation, and evaluation across additional mathematical domains to further improve performance and generalisation.

GITHUB LINK

<https://github.com/khalisahamani/AI-Duo>